

The Chinese University of Hong Kong, Shenzhen

Course Outline

2025-26 Term 2

1. Course identity

A. Course as listed at CUHK-Shenzhen

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	MDS5205
Course title (English)	Time Series Analysis
Course title (Chinese)	时间序列分析
Units	3
Description (English)	Time series data are common in finance and industry. This class introduces the related theories and applications of time series. Topics include: Trend and Seasonality, Autocovariances, Linear Time Series Analysis, Conditional Heteroscedastic Models, Nonlinear time series analysis.
Description (Chinese)	时间序列数据是在金融和工业领域中的常见数据类型。本课程介绍时间序列分析的相关理论与应用。课程主要内容包括：数据的一阶矩性质（趋势与季节性）、自协方差、线性时间序列分析、条件异方差模型以及非线性时间序列分析。

B. Corresponding course at CUHK

Please give details of the closest corresponding course at CUHK (as approved by CUHK Senate and listed in course list). If the course in Shenzhen maps to more than one course at CUHK, please make multiple copies of the block below.

Course code	STAT6104
Course title (English)	Financial Time Series
Course title (Chinese)	
Units	3
Description (English)	This course deals with the methodology and applications of business and financial time series. Topics include statistical tools useful in analysing time series, models for stationary and non-stationary time series, seasonality, forecasting techniques, heteroskedasticity, ARCH and GARCH models, and multivariate time series. (For MSc students in Data Science & Business Statistics; For MSc students in Risk Management Science)

Description (Chinese)	
-----------------------	--

2. **Course requisites**

Please state prerequisites and co-requisites, in terms of courses at CUHK-Shenzhen or any other requirements.

A. **Prerequisites**

Students should have taken basic probability and statistics courses, e.g., STA2002 or above.

B. **Co-requisites**

C. **Exclusion**

D. **Others**

3. **Learning outcomes**

1. Be familiar with concepts, theories, and frequently-used models of analyzing time series data.
2. Understand forecasting based on conditional mean and volatility.
3. Be able to implement corresponding forecasting and analysis techniques.

4. **Course syllabus**

Please highlight fundamental concepts involved in each topic to help students better understand what is covered in the course.

Trend and Seasonality: Trend-Seasonality-Stationary decomposition of time series; fitting trend and seasonality.

Autocovariances: Concept of autocovariances; autocorrelation coefficient function (acf); partial autocorrelation function (pacf); estimation of acf and pacf.

ARMA model: White noise; autoregressive moving (ARMA) process; fitting with Yule-Walker equation; prediction with ARMA process; Kalman filter.

Conditional Heteroscedastic Models: concept of Conditional heteroscedasticity; GARCH model; fitting a GARCH model.

Nonlinear time series analysis: Functional Coefficient AR Model, time-varying AR model, neural network models.

5. Assessment scheme

Component/ method	% weight
Homework	30
Midterm	30
Final project	40

Remarks:

6. Grade descriptor

The grade descriptor should align with specific course learning outcomes, not a generic description applicable to all courses.

Type: **General**

Grade	Description
A / A-	EXCELLENT - exceptionally good performance and exceeding expectations in all or most of the course learning outcomes; demonstration of a solid understanding of the subject matter, the ability to analyze problems and apply extensive knowledge, and skillful use of concepts and materials to derive proper solutions.
B+/B/B-	GOOD – good performance in all course learning outcomes and exceeding expectations in some of them; demonstration of a good understanding of the subject matter and the ability to use proper concepts and materials to solve most of the problems encountered.
C+/C/C-	FAIR – adequate performance and meeting essential expectations in course learning outcomes; demonstration of adequate understanding of the subject matter and the ability to solve simple problems.
D/D-	MARGINAL – performance barely meets expectations in the course learning outcomes; demonstration of partial understanding of the subject matter and the ability to solve simple problems.
F	FAILURE – performance does not meet expectations in the essential course learning outcomes; demonstration of serious deficiencies and the need to retake the course.

7. Feedback for evaluation

Please specify forms of evaluation to generate feedback on classes from students.

Informal feedback to the instructor and/or teaching assistant(s) through face-to-face talking, emails, office hours, etc.;

Advisor-advisee program and program review.

8. Readings

Recommended:

1. Time series: A first course with bootstrap starter., Politis, Dimitris N., and Tucker S. McElroy., Chapman and Hall/CRC, 1, 2019, United States

Others:

9. Course components

Activity	Hours/week
Homework	4
Lectures	3

10. Indicative teaching plan

Week	Content/ topic/ activity
1-2	Trend and Seasonality, Autocovariances.
3-6	ARMA model
7-10	Conditional Heteroscedastic Models
11-12	Nonlinear time series analysis
13-14	Final project

11. Implementation plan

The implementation plan may vary from year to year. Please indicate expected enrolment, and number of sections. Example: 150 students for lecture (x 2); 30 students for tutorials (x 10).

50 students for lecture (x1)

12. Academic honesty and plagiarism

A course outline may include subject-specific requirements on plagiarism, in addition to the University established policies.

Students must complete all coursework and assessments independently, unless AI tool use is explicitly permitted. Unauthorized or improper use of AI may constitute academic dishonesty and harm learning. Misuse includes submitting AI-generated work as one's own, using AI to cheat, or employing AI unethically. Always obtain permission before using AI in academic work, and be sure to acknowledge any AI use in submissions.

13. Approval

Has the course title been included in the programme submission approved by CUHK Senate? Are there any differences? Have the details (as in this document) been approved at School or other level in CUHK- Shenzhen?

Submit application

14. Any other information

15. Version date

Version number	1
As of (date)	2025-10-31